COMP3308 26s1 Final Practice

W1 Course Overview and AI Agents

Q1. Short Answer Semester 1 2024 Final (modified)
A museum wants an AI guide robot that answers visitor questions and recommends exhibits. Formulate this as an intelligent-agent problem: give a suitable PEAS description and classify two task-environment properties.

Q2. MCQ
Which option best describes a rational agent?

(A) It always imitates human behaviour.
(B) It selects actions expected to maximise its performance measure given its percepts and knowledge.
(C) It must use a neural network.
(D) It never acts under uncertainty.

Q3. True/False Pitfall
True or False: A problem formulation must include a state representation, initial state, actions, goal test, and path-cost function before search algorithms can be compared meaningfully. Justify briefly.

Q4. Short Answer
Give one example each of supervised learning, unsupervised learning, and probabilistic reasoning in a smart-library system.

Q5. Table/Diagram Strategy
Create a two-column table contrasting "thinking rationally" and "acting rationally" views of AI, including one limitation of each.

Q6. MCQ
A drone delivery environment has wind gusts and moving people. Which pair of properties is most plausible?

(A) Deterministic and static.
(B) Stochastic and dynamic.
(C) Fully observable and episodic by definition.
(D) Single-state and discrete only.

Q7. Short Answer
Why can a very accurate model still be a poor AI system if its performance measure is badly chosen? Give a concrete example.

Q8. Calculation Formula
A classifier handles 160 support tickets. It correctly routes 126 tickets, misroutes 18 urgent tickets as normal, and misroutes 16 normal tickets as urgent. Compute accuracy. Which error type may be more costly and why?

Q9. Short Answer
State two ethical or practical risks when deploying AI for medical triage. For each risk, give one mitigation.

Q10. Long Question Warning
A hospital asks for an AI triage assistant.

1. Define the task as an agent problem using PEAS.
2. Identify whether the environment is fully observable, deterministic, episodic, and static. Explain.
3. Recommend two evaluation metrics beyond accuracy.
4. Explain why AI success should be assessed after deployment, not only during training.

W2 Uninformed and Informed Search 1

Q1. Calculation Semester 1 2021 Final (modified)

Consider the graph below. Start at S, goal is G, and ties are broken alphabetically. Edge costs are shown on arcs.

$$S \to A : 1, \quad S \to B : 4, \quad S \to C : 2, \quad A \to D : 3, \quad B \to G : 2, \quad C \to E : 1, \quad E \to G : 5, \quad D \to G : 2$$

List the path returned by BFS, DFS, and UCS. State any assumptions about goal testing.

Q2. MCQ

Which algorithm is guaranteed optimal for positive, non-uniform step costs when no heuristic is available?

(A) Depth-first search.
(B) Breadth-first search.
(C) Uniform-cost search.
(D) Greedy best-first search.

Q3. True/False Pitfall

True or False: Breadth-first search is always cheaper in memory than depth-first search. Justify using asymptotic space.

Q4. Calculation Formula

A search tree has branching factor $b = 3$, shallowest goal depth $d = 5$, and maximum depth $m = 9$. Give the usual big-O time and space for BFS and DFS.

Q5. Short Answer Warning

Explain the difference between expansion order and returned path. Why can confusing them lose marks in search-trace questions?

Q6. Table/Diagram

Draw a small frontier table for BFS on the graph $S \to A, B$, $A \to C$, $B \to D$, $C \to G$, $D \to G$, assuming alphabetic tie-breaking and goal test on expansion.

Q7. MCQ

If all step costs are equal and the search tree is finite, which algorithm combines BFS optimality with DFS-like memory?

(A) Iterative deepening search.
(B) Hill climbing.
(C) Minimax.
(D) Naive Bayes.

Q8. Short Answer

Give a counterexample idea showing that UCS may expand more nodes than BFS before reaching a shallow but expensive goal.

Q9. Calculation

For paths $S{-}A{-}G$ with costs $2, 7$, $S{-}B{-}G$ with costs $4, 4$, and $S{-}C{-}D{-}G$ with costs $1, 1, 9$, which path does UCS return? Show path costs.

Q10. Long Question Strategy

A courier route planner must find a path from depot to customer locations.

1. Formulate the search problem.
2. Compare BFS, DFS, UCS, and IDS for this task.
3. State what changes if all roads have equal travel time.
4. Explain how repeated states should be handled in graph search.

W3 A* Search and Local Search

Q1. Calculation **Semester 1 2023 Final (modified)**

A graph search problem has start S, goal G, and edges $S \to A : 2$, $S \to B : 1$, $A \to G : 4$, $B \to C : 1$, $B \to G : 7$. Heuristics are $h(S) = 5, h(A) = 3, h(B) = 4, h(C) = 4, h(G) = 0$. Run A* tree search until the goal is selected for expansion. Show g, h, f values and the returned path.

Q2. MCQ

A heuristic is admissible when:

(A) It never underestimates the true remaining cost.
(B) It never overestimates the true remaining cost.
(C) It equals path cost so far.
(D) It ignores goal distance.

Q3. True/False **Pitfall**

True or False: Every consistent heuristic is admissible, assuming $h(\text{goal}) = 0$. Justify using the consistency inequality.

Q4. Calculation **Formula**

For edge $A \to B$ with cost 2, $h(A) = 6$, and $h(B) = 3$, does this edge violate consistency? Show the test.

Q5. Short Answer **Warning**

Why should A* graph search sometimes reopen a node when the heuristic is admissible but inconsistent?

Q6. Table/Diagram

Make a table with columns node, g, h, and f for a frontier containing $A : (g = 4, h = 5)$, $B : (g = 7, h = 1)$, and $C : (g = 3, h = 4)$. Which is expanded next by A*?

Q7. MCQ

A* with heuristic h_2 usually expands no more nodes than A* with h_1 when both are admissible and:

(A) $h_2(n) \geq h_1(n)$ for all n.
(B) $h_2(n) \leq h_1(n)$ for all n.
(C) h_2 is always zero.
(D) h_2 is inadmissible.

Q8. Short Answer

In simulated annealing, what happens if the temperature remains extremely high for the entire run?

Q9. Calculation

A worse move increases cost by $\Delta E = 3$. Compute its acceptance probability when $T = 1$ and when $T = 10$, leaving answers as e^{-3} and $e^{-0.3}$ if desired.

Q10. Long Question **Strategy**

A warehouse robot uses A* with a grid heuristic.

1. Propose an admissible heuristic for four-direction grid movement.
2. Explain admissibility and consistency for your heuristic.
3. Explain one case where greedy best-first search may be faster but worse.
4. Describe when local search would be inappropriate for route planning.

W4 Game Playing, Minimax and Alpha-Beta

Q1. Calculation Semester 1 2022 Final (modified)

A complete two-ply game tree has MAX at root, MIN children B, C, and leaf values under B: $6, 2, 5$; under C: $1,$ are visited left to right. Compute minimax values, the selected move, and any alpha-beta pruning.

Q2. MCQ

Alpha-beta pruning:

(A) Changes the minimax value to a heuristic value.
(B) Returns the same minimax value while avoiding irrelevant branches.
(C) Works only for stochastic games.
(D) Removes the need for an evaluation function at cut-off depth.

Q3. True/False Pitfall

True or False: Good move ordering can increase the amount of alpha-beta pruning. Justify.

Q4. Calculation Formula

At a MIN node, current $\beta = 5$. An ancestor MAX node has $\alpha = 7$. Should the remaining children of this MIN node be pruned? Explain.

Q5. Short Answer Warning

What is the horizon effect in depth-limited game search? Give one mitigation.

Q6. Table/Diagram

Draw a minimax value table for a tree where root A is MAX, B, C are MIN nodes, D, E, F, G are MAX nodes, and leaf pairs are $D : (4, 9)$, $E : (3, 6)$, $F : (2, 8)$, $G : (1, 5)$.

Q7. MCQ

In a zero-sum deterministic game with perfect information, the opponent at a MIN node is assumed to:

(A) Choose randomly.
(B) Minimise MAX's utility.
(C) Maximise MAX's utility.
(D) Ignore utilities.

Q8. Short Answer

Why does alpha-beta pruning depend on the order in which children are visited, even though the final minimax value does not?

Q9. Calculation

For a game tree with branching factor $b = 4$ and depth $m = 6$, give the usual minimax time complexity and the ideal alpha-beta effective complexity.

Q10. Long Question Strategy

A chess-like game agent must move within 2 seconds.

1. Explain why exhaustive search is infeasible.
2. Describe minimax with a depth cutoff and evaluation function.
3. Explain alpha-beta pruning and the role of move ordering.
4. State one limitation of using a fixed-depth cutoff.

W5 Machine Learning, KNN and Rules

Q1. Short Answer **Semester 1 2021 Final (modified)**
A bank asks for an explainable classifier for loan approvals. Compare KNN, 1R/rule learning, and decision trees for
Mention interpretability and prediction-time cost.

Q2. MCQ
K-nearest neighbours is usually described as:

(A) Lazy and instance-based.
(B) Eager and parametric.
(C) Unsupervised only.
(D) A probabilistic graphical model.

Q3. True/False **Pitfall**
True or False: Feature normalisation is often important for KNN with Euclidean distance. Justify.

Q4. Calculation **Formula**
Training points are $A(0,0)$ yes, $B(2,0)$ yes, $C(4,0)$ no, $D(5,0)$ no. Classify $x = (3,0)$ using 1-NN and 3-NN. For 1-NN ties, choose the alphabetically earlier point ID; for 3-NN class ties, choose the class with smaller total neighbour distance.

Q5. Short Answer **Warning**
Why can KNN perform poorly in high-dimensional data? Name one remedy.

Q6. Table/Diagram
Construct a table comparing supervised, unsupervised, and reinforcement learning, with one COMP3308-style example for each.

Q7. MCQ
Which statement about 1R is most accurate?

(A) It builds one-rule classifiers using a single attribute.
(B) It always beats random forests.
(C) It cannot produce interpretable output.
(D) It is the same as A*.

Q8. Short Answer
Explain coverage and accuracy/confidence of an IF-THEN rule.

Q9. Calculation
A rule covers 40 examples; 30 are positive and 10 are negative. In a dataset of 200 examples, compute coverage fraction and rule accuracy for the positive class.

Q10. Long Question **Strategy**
A university wants to classify whether students need early academic support.

1. Identify the learning task and target label.
2. Compare KNN and rule-based learning for this task.
3. Explain what validation data is used for.
4. State two risks of using historical student data.

W6 Naive Bayes and Evaluation

Q1. Calculation **Semester 1 2022 Final (modified)**

A small training set predicts $Like \in \{yes, no\}$ from genre and country:

$rock$	AU	yes
$jazz$	AU	yes
$jazz$	UK	no
$folk$	UK	no
$rock$	US	no
$jazz$	US	yes

Use Naive Bayes without smoothing to classify $genre = jazz, country = AU$. Show both class scores.

Q2. MCQ

Naive Bayes assumes that features are:

(A) Marginally independent.
(B) Conditionally independent given the class.
(C) Always numeric.
(D) Always normally distributed.

Q3. True/False **Pitfall**

True or False: The denominator $P(x)$ in Bayes' theorem is unnecessary when only comparing class labels in Naive Bayes. Justify.

Q4. Calculation **Formula**

There are 12 spam and 18 non-spam emails. The word "offer" appears in 6 spam emails and 3 non-spam emails. Compute unsmoothed $P(spam)$, $P(nonspam)$, $P(offer|spam)$, and $P(offer|nonspam)$.

Q5. Short Answer **Warning**

What is the zero-frequency problem? How does Laplace smoothing address it?

Q6. Table/Diagram

Draw a confusion matrix and label TP, FP, FN, and TN for positive class "high risk".

Q7. MCQ

If false negatives are much more costly than false positives, which metric is usually emphasised?

(A) Recall/sensitivity.
(B) File size.
(C) Number of attributes.
(D) Training set order.

Q8. Calculation

A model has TP=42, FP=8, FN=18, TN=132. Compute accuracy, precision, recall, and F_1.

Q9. Short Answer

Explain why accuracy can be misleading for highly imbalanced classes.

Q10. Long Question **Strategy**

A news app uses Naive Bayes to predict whether a user clicks an article.

1. Write the Naive Bayes scoring formula for two features.
2. Explain how to estimate priors and likelihoods from training data.
3. Explain how to handle an unseen feature value.
4. Choose two evaluation metrics and justify them.

W7 Decision Trees

Q1. Calculation Semester 1 2023 Final (modified)
A dataset has six examples with class $Risk$: two high and four low. Splitting by Season gives Summer: two h
Winter: two low; Autumn: one low. Compute $H(Risk)$, $H(Risk|Season)$, and $IG(Risk, Season)$.

Q2. MCQ
Information gain in decision-tree learning measures:

(A) The reduction in impurity after a split.
(B) The number of leaves only.
(C) The Euclidean distance between examples.
(D) The SVM margin.

Q3. True/False Pitfall
True or False: A fully grown decision tree can be sensitive to noise and may overfit. Justify.

Q4. Calculation Formula
Compute entropy for a node containing 3 positive and 3 negative examples. Then compute entropy for a pure node containing 6 positive examples.

Q5. Short Answer
What is the difference between pre-pruning and post-pruning?

Q6. Table/Diagram
Draw a tiny decision tree with root attribute Weather and leaves for at least two branches. State the corresponding IF-THEN rules.

Q7. MCQ
A decision tree with axis-aligned numeric splits most naturally creates:

(A) Rectangular decision regions in feature space.
(B) Only circular clusters.
(C) A Bayesian network.
(D) A minimax tree.

Q8. Short Answer Warning
Why can information gain be biased toward attributes with many values? Name one alternative measure.

Q9. Calculation
Attribute A splits 10 examples into S_1 with entropy 0.5 and 4 examples, and S_2 with entropy 1.0 and 6 examples. If parent entropy is 0.971, compute information gain.

Q10. Long Question Strategy
A decision tree is trained to approve scholarship applications.

1. Describe how the root split is selected using entropy and information gain.
2. Explain how numeric attributes can be handled.
3. Explain why pruning may improve test performance.
4. Discuss one interpretability advantage and one fairness risk.

W8 Perceptrons and Neural Networks 1

Q1. Calculation **Semester 1 2021 Final (modified)**
A perceptron has $w_1 = 0.2, w_2 = -0.3, b = 0.1$, learning rate $\eta = 0.5$, and step output 1 if $n \geq 0$. For trai
$x = (1, -1)$, desired output $t = 0$, compute output, error, and updated weights and bias.

Q2. MCQ
A single perceptron with a step activation can solve:

(A) Only linearly separable binary classification problems.
(B) Every possible classification problem.
(C) Only clustering problems.
(D) Bayesian-network inference.

Q3. True/False **Pitfall**
True or False: A neural network with step transfer functions is suitable for standard backpropagation training. Justify.

Q4. Calculation **Formula**
For $w = (0.4, -0.1)$, $b = 0.2$, $x = (0.6, 0.8)$, compute $n = w^\top x + b$ and the step output.

Q5. Short Answer **Warning**
Why should hidden-layer weights not all be initialised to identical values?

Q6. Table/Diagram
Draw a small neural network diagram with two inputs, two hidden neurons, and one output. Label weights conceptually.

Q7. MCQ
During classification with a trained feed-forward neural network, the main operation is:

(A) Forward propagation of activations.
(B) Backward propagation and weight update.
(C) Alpha-beta pruning.
(D) Agglomerative clustering.

Q8. Short Answer
State one reason why a multilayer perceptron can solve XOR but a single perceptron cannot.

Q9. Calculation
If a neuron uses sigmoid activation $\sigma(z) = 1/(1+e^{-z})$, leave the output for $z = 0.7$ in exact exponential form and state whether it is greater than 0.5.

Q10. Long Question **Strategy**
A small neural network is proposed for fraud detection.

1. Explain the role of input, hidden, and output layers.
2. Describe one perceptron update step.
3. Explain why differentiable activations matter for backpropagation.
4. State two signs of overfitting during training.

W9 Multilayer Neural Networks and Deep Learning

Q1. Calculation **Semester 1 2024 Final (modified)**
An input image has size 6×5 with one channel. A valid convolution uses a 3×2 filter, stride 1, no padding. Compute the size of each feature map and the total number of convolution-layer neurons.

Q2. MCQ
Without nonlinear activation functions, a multilayer neural network is equivalent to:

(A) A single linear transformation.
(B) A k-means model.
(C) A random forest.
(D) An alpha-beta search.

Q3. True/False **Pitfall**
True or False: Backpropagation is guaranteed to converge to the same weights from every random initialisation. Justify.

Q4. Calculation **Formula**
For a 4×4 input patch and filter

$$X = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 2 & 1 & 0 \\ 2 & 1 & 0 & 2 \\ 1 & 1 & 2 & 0 \end{bmatrix}, \quad F = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix},$$

compute the first row of the valid convolution output.

Q5. Short Answer **Novel**
How can an autoencoder be used to improve or pretrain a deep neural network?

Q6. Table/Diagram
Draw a pipeline diagram for image classification using convolution, activation, pooling, flattening, and dense classification.

Q7. MCQ
Which symptom most directly suggests overfitting?

(A) Training loss decreases while validation loss rises.
(B) Both training and validation loss remain very high.
(C) Learning rate is exactly zero.
(D) The dataset has labels.

Q8. Short Answer
Compare batch gradient descent and stochastic/mini-batch gradient descent.

Q9. Calculation
A network has input dimension 20, one hidden layer with 5 neurons, and one output neuron. Count the trainable parameters, including biases, for fully connected layers.

Q10. Long Question **Strategy**
A company wants deep learning for image-based defect detection.

1. Explain why CNNs are suitable for images.
2. Compute valid convolution output dimensions in general for $N \times M$ input and $A \times B$ filter.
3. Give two regularisation methods.
4. Explain why a held-out validation set is needed.

W10 Support Vector Machines and Ensembles

Q1. Short Answer **Semester 1 2021 Final (modified)**

A statement claims: "The SVM kernel trick guarantees that any dataset becomes perfectly linearly separable." D
this is true and explain the role of soft margin and kernels.

Q2. MCQ

The support vectors in an SVM are:

(A) Training points that determine the margin/boundary.
(B) All points not used in training.
(C) Random forest trees.
(D) Bayesian-network parents.

Q3. True/False **Pitfall**

True or False: A basic perceptron and a linear hard-margin SVM always learn the same separating hyperplane on separable data. Justify.

Q4. Calculation **Formula**

For $w = (3, 4)$, $b = -5$, compute the geometric margin width $2/\|w\|$. Classify $x = (2, 0)$ by sign of $w^\top x + b$.

Q5. Short Answer **Warning**

Explain the effect of increasing C in a soft-margin SVM.

Q6. Table/Diagram

Draw a small maximum-margin diagram with a separating line, two margin lines, and support vectors.

Q7. MCQ

Random forests inject randomness mainly by:

(A) Bootstrap samples and random feature subsets at splits.
(B) Replacing labels with random labels.
(C) Never using decision trees.
(D) Running A* repeatedly.

Q8. Short Answer

Contrast bagging and boosting in terms of training order and how errors are handled.

Q9. Calculation

An AdaBoost weak learner has weighted error $\epsilon = 0.2$. Compute $\alpha = \frac{1}{2}\ln((1-\epsilon)/\epsilon)$. Leave as a logarithm or approximate qualitatively as positive.

Q10. Long Question **Strategy**

A regulator asks for a robust classifier for credit decisions.

1. Compare linear SVM, kernel SVM, single decision tree, and random forest.
2. Discuss interpretability versus predictive performance.
3. Explain how false-positive and false-negative costs affect model choice.
4. Recommend one model and justify assumptions.

W11 Probabilistic Reasoning and Bayesian Networks

Q1. Calculation Semester 1 2024 Final (modified)
Consider the Bayesian network $O \to C$, $O \to Q$, $A \to Q$, $C \to G$, $Q \to G$. Write the joint factorisation. $P(q, c, g)$ as a summation over hidden variables.

Q2. MCQ
In a Bayesian network, the full joint distribution factorises as:

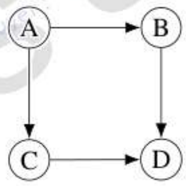
(A) $\prod_i P(X_i|\text{Parents}(X_i))$.
(B) $\sum_i P(X_i)$.
(C) The maximum of all CPT entries.
(D) A k-means objective.

Q3. True/False Pitfall
True or False: Observing a collider can make its parent variables dependent. Justify with a short example.

Q4. Calculation Formula
Given $P(R) = 0.2$, $P(S|R) = 0.8$, $P(S|\neg R) = 0.1$, compute $P(S)$ by total probability.

Q5. Short Answer Warning
Explain the difference between a chain, a fork, and a collider for d-separation.

Q6. Table/Diagram
Draw the following Bayesian network in TikZ or as an ASCII diagram, and state one conditional independence:



Q7. MCQ
Variable elimination can be faster than full enumeration because it:

(A) Multiplies and sums factors locally, avoiding repeated work.
(B) Ignores all hidden variables.
(C) Requires no CPTs.
(D) Converts the BN into KNN.

Q8. Short Answer
How is Naive Bayes represented as a Bayesian network?

Q9. Calculation
If $P(A) = 0.3$, $P(B) = 0.4$, and A, B are independent parents of C, with $P(c|a, b) = 0.9$, $P(c|a, \neg b) = 0.5$, $P(c|\neg a, b) = 0.4$, $P(c|\neg a, \neg b) = 0.1$, compute $P(c)$.

Q10. Long Question Strategy
A disease diagnosis BN has Disease $\to$ Test and Disease $\to$ Symptom.

1. Draw the network and write its factorisation.
2. Explain whether Test and Symptom are independent marginally and conditionally on Disease.
3. Show how to compute $P(Disease|Test, Symptom)$ up to proportionality.
4. State one data issue when learning CPTs.

W12 Unsupervised Learning and Clustering

Q1. Calculation **Semester 1 2022 Final (modified)**
Items A, B, C, D, E have distances $d(A,B) = 1$, $d(A,C) = 5$, $d(A,D) = 4$, $d(A,E) = 4$, $d(B,C) = 6$, d [illegible] $d(B,E) = 2$, $d(C,D) = 1$, $d(C,E) = 5$, $d(D,E) = 7$. Run single-link nearest-neighbour agglomerative cl[illegible] threshold 3. Show merges and final clusters.

Q2. MCQ
K-means is most likely to struggle when clusters are:

(A) Roughly spherical and similarly sized.
(B) Elongated, non-convex, or strongly affected by outliers.
(C) Clearly separated by Euclidean distance.
(D) Pre-scaled numeric data.

Q3. True/False **Pitfall**
True or False: Increasing k in k-means always produces a more meaningful clustering. Justify.

Q4. Calculation **Formula**
Points $0, 2, 8, 10$ on a line are clustered by k-means with $k = 2$, initial centroids 0 and 10. Perform one assignment step and one centroid update.

Q5. Short Answer **Warning**
Why should distance-based clustering usually scale numeric features before clustering?

Q6. Table/Diagram
Create a table comparing k-means, single-link hierarchical clustering, and complete-link hierarchical clustering.

Q7. MCQ
Single-link agglomerative clustering defines distance between two clusters as:

(A) The smallest pairwise distance between their points.
(B) The largest pairwise distance only.
(C) The product of all coordinates.
(D) The SVM margin.

Q8. Short Answer
Explain the "chaining" effect in single-link clustering.

Q9. Calculation
A dataset has 2^{30} vectors, each with 12 components stored in 4 bytes. It is compressed by replacing each vector with the index of one of 2^{16} centroids and storing all centroids. Write the total compressed bytes formula using minimum index bits.

Q10. Long Question **Strategy**
A wildlife project clusters animal movement tracks.

1. Compare k-means and hierarchical clustering for this task.
2. Explain how to choose or validate the number of clusters.
3. Discuss one limitation of using Euclidean distance on raw features.
4. Recommend a clustering workflow from preprocessing to interpretation.

Answer Key

W1

Q1. PEAS: performance = helpful and safe recommendations; environment = museum, visitors, exhibits; actuators = speech, display, wheels; sensors = camera, mic, location. Partially observable and dynamic/stochastic are defensible.
Q2. B.
Q3. True. Without those components the algorithms may solve different problems or use different cost notions.
Q4. Supervised: predict overdue book risk from labelled history; unsupervised: cluster borrowing patterns; probabilistic reasoning: infer likely user intent from noisy query evidence.
Q5. Thinking rationally uses formal logic but can be brittle/intractable; acting rationally chooses best action under uncertainty but depends on performance measure and model.
Q6. B.
Q7. A wrong performance measure optimises the wrong behaviour, e.g. maximising triage throughput may neglect severe cases.
Q8. Accuracy $= 126/160 = 0.7875$. Misrouting urgent as normal is likely costlier because needed intervention is delayed.
Q9. Bias/unfairness mitigated by auditing and representative data; privacy mitigated by minimisation/access controls; unsafe recommendations mitigated by human review.
Q10. Strong answers define PEAS, classify partial observability/stochastic/sequential/dynamic, choose recall/sensitivity and precision or F_β, and explain monitoring for drift, real-world cost, and stakeholder acceptance.

W2

Q1. With goal test on expansion: BFS returns $S-B-G$. DFS alphabetic usually returns $S-A-D-G$. UCS returns cost 6; alphabetic tie handling gives $S-B-G$ before the equal-cost $S-A-D-G$.
Q2. C.
Q3. False. BFS stores frontier $O(b^{d+1})$; DFS stores $O(bm)$.
Q4. BFS time/space $O(3^6)$ style $O(b^{d+1})$; DFS time $O(3^9)$, space $O(3 \cdot 9)$.
Q5. Expansion order is nodes removed/expanded from frontier; returned path is parent chain to the goal.
Q6. BFS frontier evolves $[S] \to [A,B] \to [B,C] \to [C,D] \to [D,G] \to [G]$; expand S, A, B, C, D, G if testing on expansion.
Q7. A.
Q8. Example: a depth-1 goal cost 100 and another branch containing many cost-1 nodes; UCS explores cheap branch first.
Q9. Costs: $SAG = 9$, $SBG = 8$, $SCDG = 11$; UCS returns $S-B-G$.
Q10. Complete answer defines states/actions/goal/cost, compares completeness/optimality/memory, notes BFS/IDS if equal costs, UCS for positive nonuniform costs, and explored/best-g table for repeated states.

W3

Q1. Start $S : f = 5$. Expand S: $A : g2, h3, f5$, $B : g1, h4, f5$; alphabetic expands A, generate $G : g6, f6$; expand B, generate $C : g2, h4, f6$, $G : g8$; expand C or G by tie rule. Best goal is $S-A-G$ cost 6; if C expanded first it generates $G : g7$.
Q2. B.
Q3. True. By applying $h(n) \le c + h(n')$ along a path to a goal.
Q4. Need $6 \le 2 + 3 = 5$, false, so inconsistent on this edge.
Q5. Because a later path may give a lower g; without reopening, graph search can lock in a suboptimal path.
Q6. $A : f9$, $B : f8$, $C : f7$; expand C.
Q7. A.
Q8. It behaves almost randomly for worse moves because even bad moves are accepted with high probability.
Q9. $T = 1 : e^{-3}$; $T = 10 : e^{-0.3}$, which is larger.
Q10. Manhattan distance is admissible/consistent for four-direction unit grid without negative costs; greedy uses only h and may choose a dead-end; local search unsuitable when exact path and optimality matter.

W4

Q1. $B = \min(6,2,5) = 2$, $C = \min(1,8,4) = 1$, root $= 2$, choose B. With left-to-right, after $B = 2$, at C first child 1 gives $\beta = 1 \le \alpha = 2$, prune C's remaining leaves.
Q2. B.
Q3. True. Earlier good MAX moves raise α, earlier good MIN moves lower β, creating more cutoffs.
Q4. Yes. At MIN, $\beta = 5 \le \alpha = 7$, so prune remaining children.
Q5. Cutoff misses consequences just beyond depth; use quiescence search or deeper selective search.
Q6. $D = 9, E = 6, F = 8, G = 5$; then $B = \min(9,6) = 6$, $C = \min(8,5) = 5$, root $A = 6$.

Q7. B.
Q8. Ordering changes when α, β bounds become tight, but minimax value is mathematically unchanged.
Q9. Minimax $O(4^6)$; ideal alpha-beta about $O(4^3)$.
Q10. Strong answer covers branching explosion, heuristic evaluation at cutoff, alpha-beta bounds/pruning, move horizon effect.

W5

Q1. KNN can cite neighbours but prediction is slow and scale-sensitive; 1R is very interpretable but weak; trees give clear rules and better interactions, so often best for explainable loan decisions with validation.
Q2. A.
Q3. True. Large-scale features dominate Euclidean distance unless scaled.
Q4. 1-NN has B and C tied at distance 1, so alphabetic point tie gives B = yes. For 3-NN, nearest B(yes), C(no), D(no), so predict no.
Q5. Distances become less discriminative; remedy feature selection, PCA, scaling, or alternative model.
Q6. Supervised uses labels, unsupervised clusters, reinforcement learns from rewards.
Q7. A.
Q8. Coverage is matched examples; accuracy/confidence is correct matched examples divided by matched examples.
Q9. Coverage fraction $40/200 = 0.2$; accuracy $30/40 = 0.75$.
Q10. Strong answer identifies binary classification target, compares KNN/rules on interpretability, scaling, update and high dimension, uses validation for hyperparameters, and notes bias/privacy risks.

W6

Q1. Priors yes $= 3/6$, no $= 3/6$. yes score $= 3/6 \cdot 2/3 \cdot 2/3 = 2/9$. no score $= 3/6 \cdot 1/3 \cdot 0/3 = 0$. Predict yes.
Q2. B.
Q3. True. $P(x)$ is common to all classes for the same example.
Q4. $P(spam) = 12/30$, $P(nonspam) = 18/30$, $P(offer|spam) = 6/12$, $P(offer|nonspam) = 3/18$.
Q5. Zero counts make a class score zero; Laplace adds pseudo-counts, e.g. $(count + 1)/(n_C + m)$.
Q6. Matrix rows actual, columns predicted: TP high/high, FN high/low, FP low/high, TN low/low.
Q7. A.
Q8. Accuracy $174/200 = 0.87$; precision $42/50 = 0.84$; recall $42/60 = 0.70$; $F_1 = 2(0.84)(0.70)/(1.54) \approx 0.764$.
Q9. A majority-class classifier can have high accuracy while missing rare positives.
Q10. Strong answer gives $P(C)P(x_1|C)P(x_2|C)$, estimates from counts, applies smoothing, and selects metrics based on business costs and imbalance.

W7

Q1. $H = 0.918$. Conditional entropy $= 3/6(0.918) + 2/6(0) + 1/6(0) = 0.459$. IG $= 0.459$.
Q2. A.
Q3. True. Full trees can memorise noise; pruning can improve generalisation.
Q4. $H(3,3) = 1$; pure entropy 0.
Q5. Pre-pruning stops growth early; post-pruning grows then removes branches using validation/error estimate.
Q6. Example: Weather=sunny -> no, overcast -> yes, rainy -> test Wind. Rules are root-to-leaf paths.
Q7. A.
Q8. Many-valued attrs can isolate examples and give high gain; gain ratio is an alternative.
Q9. Weighted entropy $= 4/10(0.5) + 6/10(1) = 0.8$; gain $= 0.971 - 0.8 = 0.171$.
Q10. Strong answer shows entropy/IG split choice, threshold candidates for numeric attrs, pruning to reduce variance, interpretability plus fairness risk from biased features.

W8

Q1. $n = 0.2(1) + (-0.3)(-1) + 0.1 = 0.6$, $y = 1$, $e = -1$. Updates: $w_1 = 0.2 - 0.5 = -0.3$, $w_2 = -0.3 + 0.5 = 0.2$, $b = 0.1 - 0.5 = -0.4$.
Q2. A.
Q3. False. Step is not differentiable, so standard backprop cannot use its gradient.
Q4. $n = 0.4(0.6) - 0.1(0.8) + 0.2 = 0.36$, output 1.
Q5. Identical hidden units get identical gradients and remain redundant; random init breaks symmetry.

Q6. Any correct two-input, two-hidden, one-output feed-forward diagram with conceptual weights earns marks.
Q7. A.
Q8. Hidden nonlinear layers combine linear boundaries to form nonlinear decision regions such as XOR.
Q9. $\sigma(0.7) = 1/(1 + e^{-0.7}) > 0.5$.
Q10. Strong answer explains layers, perceptron update $w \leftarrow w + \eta e x$, differentiability for gradients, and overfitti[illegible] as low train loss/high validation loss.

W9

Q1. Each map size $(6 - 3 + 1) \times (5 - 2 + 1) = 4 \times 4$. With 4 filters: $4 \cdot 4 \cdot 4 = 64$ neurons.
Q2. A.
Q3. False. Nonconvex optimisation and random initialisation can lead to different weights/local minima.
Q4. First row patches: $1(1) + 0(-1) + 0(0) + 2(2) = 5$; $0(1) + 2(-1) + 2(0) + 1(2) = 0$; $2(1) + 1(-1) + 1(0) + 0(2) = 1$.
Q5. Autoencoders learn compact/denoised representations; encoder weights can initialise a classifier or reduce dimensionality.
Q6. Conv -> activation -> pooling -> flatten -> dense -> softmax/class output.
Q7. A.
Q8. Batch uses all data per update, stable but expensive; SGD/mini-batch uses noisy smaller updates, faster and often generalises well.
Q9. Input-hidden: $20 \cdot 5 + 5 = 105$. Hidden-output: $5 \cdot 1 + 1 = 6$. Total 111.
Q10. Strong answer mentions local receptive fields, weight sharing, valid output $(N - A + 1)(M - B + 1)$, regularisation such as dropout/early stopping/augmentation, and validation for tuning/overfit control.

W10

Q1. False. Kernels compute inner products in feature space; separability is not guaranteed. Soft margin permits violations and trades margin against errors.
Q2. A.
Q3. False. A perceptron may find any separator depending on order; hard-margin SVM finds the maximum-margin separator.
Q4. $\|w\| = 5$, margin width $2/5 = 0.4$. $w^\top x + b = 6 - 5 = 1$, positive.
Q5. Larger C penalises violations more, usually narrower margin and lower training error but more overfit risk.
Q6. Diagram should show separator, parallel margins, closest points labelled support vectors.
Q7. A.
Q8. Bagging trains base learners in parallel on bootstrap samples; boosting trains sequentially and focuses later learners on previous mistakes.
Q9. $\alpha = \frac{1}{2}\ln(0.8/0.2) = \frac{1}{2}\ln 4 \approx 0.693 > 0$.
Q10. Strong answer compares max-margin, kernels, tree interpretability, RF robustness, error costs, and justifies a model under regulatory needs.

W11

Q1. Factorisation: $P(O, A, C, Q, G) = P(O)P(A)P(C|O)P(Q|O, A)P(G|C, Q)$. Then $P(q, c, g)$ sums this product over hidden o and a.
Q2. A.
Q3. True. In $A \to C \leftarrow B$, observing C can make A and B dependent via explaining away.
Q4. $P(S) = 0.8(0.2) + 0.1(0.8) = 0.24$.
Q5. Chain $A \to B \to C$, fork $A \leftarrow B \to C$, collider $A \to B \leftarrow C$; observed middle blocks chain/fork but opens collider.
Q6. Factorisation $P(A)P(B|A)P(C|A)P(D|B, C)$. Example independence: $B \perp C|A$ if no other active path.
Q7. A.
Q8. Class node is parent of each feature node; features are conditionally independent given class.
Q9. $0.3 \cdot 0.4 \cdot 0.9 + 0.3 \cdot 0.6 \cdot 0.5 + 0.7 \cdot 0.4 \cdot 0.4 + 0.7 \cdot 0.6 \cdot 0.1 = 0.352$.
Q10. Strong answer draws $D \to T, D \to S$, factorises $P(D)P(T|D)P(S|D)$, says T and S dependent marginally but independent given D, and computes posterior proportional to $P(D)P(T|D)P(S|D)$.

W12

Q1. Merge A, B at 1; merge C, D at 1; merge E with $\{A, B\}$ at single-link distance 2. Stop because next smallest between clusters exceeds 3. Final $\{A, B, E\}, \{C, D\}$.
Q2. B.

Q3. False. Larger k lowers SSE but may split natural groups and reduce interpretability.
Q4. Assign 0,2 to centroid 0 and 8,10 to centroid 10. Updated centroids are 1 and 9.
Q5. Large-scale features dominate distance even if not more meaningful.
Q6. K-means needs k and assumes centroid-shaped clusters; single-link can chain; complete-link makes compact
Q7. A.
Q8. Single-link can connect clusters through a chain of close intermediate points, forming elongated clusters.
Q9. Total $= 2^{30} \cdot 16/8 + 2^{16} \cdot 12 \cdot 4$ bytes.
Q10. Strong answer compares algorithms, chooses cluster validation (elbow/silhouette/domain), scales/features distances, and proposes preprocessing, clustering, validation, and interpretation.